THAPAR INSTITUTE OF ENGINEERING AND TECHNOLOGY, PATIALA, PUNJAB

THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

Computer Science and Engineering Department

Database Management Systems (UCS310)

Project synopsis

Industrial Pollution Monitoring & Compliance Management System

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Submitted To:

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Introduction

In many industrial areas, environmental pollution data such as air quality, water contamination, and noise levels is recorded manually or stored in separate systems. Because these records are not centralized, it becomes difficult for environmental authorities to monitor pollution levels, track industrial compliance, and identify violations in time.

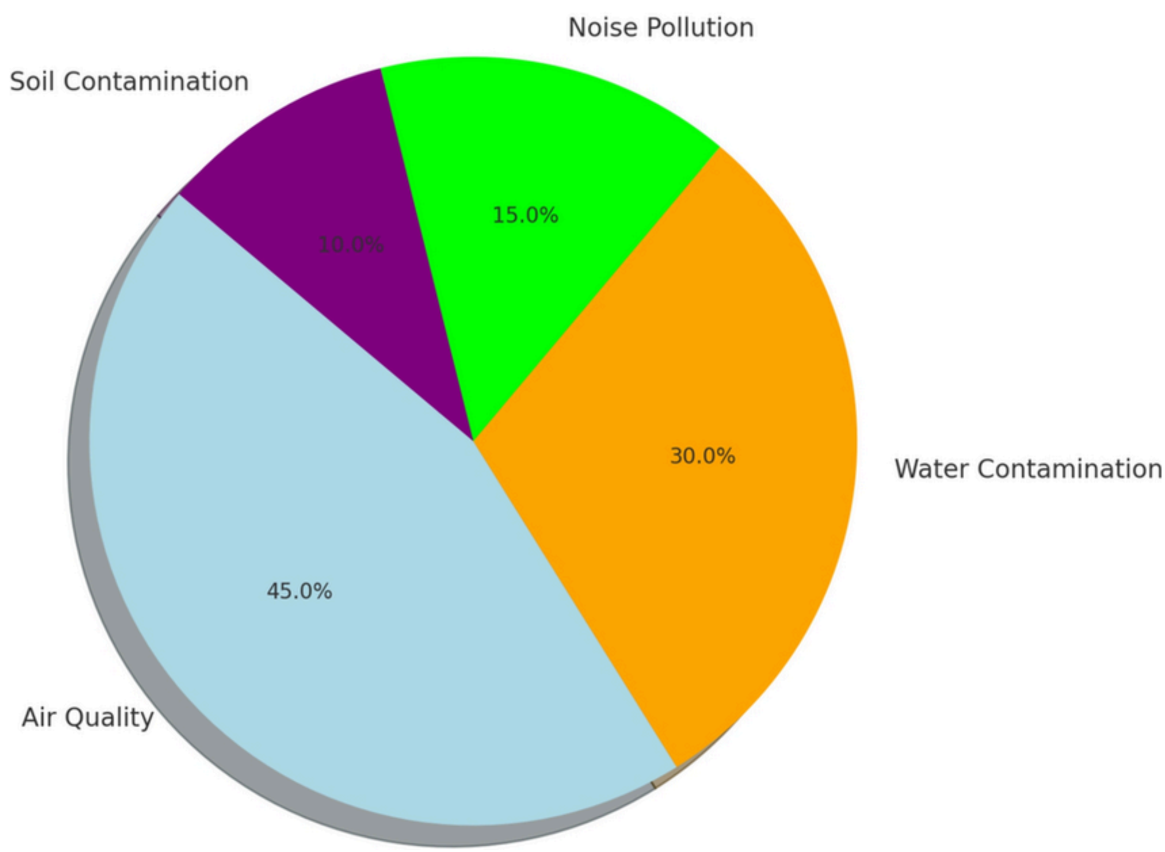
A Database Management System (DBMS) can help solve this problem by storing pollution monitoring data in a structured and organized way. It ensures accurate record-keeping, reduces redundancy, and allows quick retrieval of environmental information. Unlike file-based systems, a DBMS provides better data consistency, security, and support for automated monitoring. This project focuses on designing a centralized pollution monitoring database for industrial areas that integrates pollution readings, inspection records, and compliance tracking.

Problem Statement

Environmental pollution in industrial zones is monitored through different sources such as sensors, inspections, and reports, but these records are often not integrated into a single system.

This creates several problems:

- Pollution readings are difficult to track over time.
- Compliance violations are not recorded systematically.
- Environmental inspections are hard to manage.
- Pollution trend analysis becomes difficult.
- Authorities cannot easily identify high-pollution industries.



Because of these issues, pollution monitoring becomes inefficient and delayed actions may increase environmental damage. A centralized database system can store pollution parameters, inspection data, and compliance records in one place, improving monitoring and decision-making.

Objectives of the Project

1. To design a database using ER modeling for industrial pollution monitoring.
2. To convert the ER model into relational tables.
3. To apply normalization up to 3NF.
4. To implement pollution monitoring using SQL queries.
5. To use PL/SQL components such as triggers, procedures, and functions.
6. To maintain data integrity using constraints and transactions.

Scope of the Project

This project focuses on designing a backend DBMS-based pollution monitoring system for industrial areas.

Functional Boundaries

- Register industries and monitoring stations
- Record pollution readings (air, water, noise).
- Track inspection records.
- Monitor compliance violations.
- Generate pollution reports.

Users

- Environmental authority (Admin)
- Inspectors
- Industry representatives

Modules

- Industry management
- Monitoring station management
- Pollution data recording
- Inspection management
- Compliance tracking
- Reporting

Proposed System Description

The proposed system is a centralized pollution monitoring database that records environmental pollution parameters from industrial areas.

Working:

- Industries and monitoring stations are registered.
- Pollution readings are recorded periodically.
- Inspection reports are stored in the database.
- When pollution exceeds allowed limits, violations are recorded.
- Authorities can generate reports to monitor pollution trends.

The system improves efficiency by providing organized environmental records and automated compliance tracking. Data integrity is maintained using primary keys, foreign keys, and constraints.

Database Design

Entities

- Industry
- Location
- MonitoringStation
- PollutionReading
- Inspection
- Violation

Attributes:

1. Industry Table
industry_id (PK), industry_name, industry_type, license_number, location_id (FK)

2. Location Table
location_id (PK), area_name, city

3. MonitoringStation Table
station_id (PK), location_id (FK), station_type

4. PollutionReading Table
reading_id (PK), station_id (FK), reading_datetime, PM25, PM10, NO2, SO2, water_ph, noise_level

5. Inspection Table
inspection_id (PK), industry_id (FK), inspection_date, inspector_name, remarks, result

6. Violation Table
violation_id (PK), industry_id (FK), reading_id (FK), violation_type, penalty_amount, status

Relationships:

- One Location can have multiple Industries.
- One Location can have multiple Monitoring Stations.
- One Monitoring Station can have multiple Pollution Readings.
- One Industry can have multiple Inspections.
- One Industry can have multiple Violations.
- One PollutionReading can be associated with multiple Violations

Cardinality & Constraints:

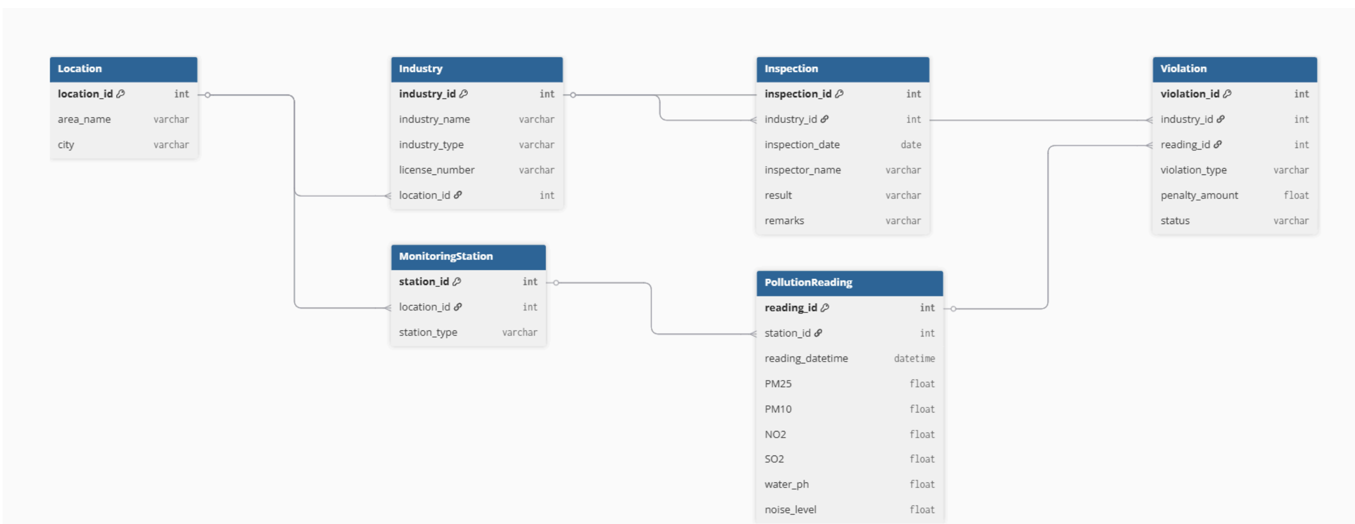
One-to-Many relationships between core entities

Primary keys uniquely identify records

Foreign keys maintain referential integrity

Constraints ensure valid pollution values

Structure:



Normalization

The database will be normalized up to Third Normal Form (3NF) to eliminate redundancy and maintain consistency in pollution monitoring records.

Database Implementation

SQL

- CREATE, INSERT, UPDATE, DELETE
- Joins and subqueries
- Aggregate queries
- Views for pollution reports

PL/SQL

- Procedure for generating pollution reports
- Function to calculate pollution severity
- Trigger for compliance violation detection
- Cursor for inspection reporting
- Exception handling

Transaction Management

Transactions using COMMIT and ROLLBACK will ensure reliable storage and ACID properties.

Tools & Technologies

MySQL / Oracle / PostgreSQL

SQL and PL/SQL

MySQL Workbench / SQL Developer

Expected Outcomes

Centralized pollution monitoring database

Efficient pollution data retrieval

Automated compliance tracking

Improved environmental monitoring

Accurate pollution trend reporting

Conclusion

This project demonstrates how a Database Management System can be used to manage environmental pollution monitoring in industrial areas. By storing pollution readings, inspection records, and compliance data in a centralized system, environmental authorities can monitor pollution levels more effectively. From a DBMS perspective, the project demonstrates database design, normalization, SQL implementation, and PL/SQL automation for real-world environmental monitoring.